Binary Content Protocol (BCP) - version 1

Wire specification | TCP port 9000 | all multi-byte integers unsigned, big-endian

1. Connection model

BCP retrieves named files. A client opens one TCP connection and may make several exchanges on it. Version 1 sends one request frame, reads its response, then sends the next request. Responses echo the request's nonzero stream ID. TCP has no message boundaries: read exactly 12 fixed-header bytes, then exactly the stated payload length. A short read is normal; EOF inside a frame is an error.

2. Fixed frame header - 12 bytes

Offset	Bytes	Field	Value or meaning
0	2	magic	ASCII BC (hex 42 43)
2	1	version	01
3	1	type	01 request; 02 response
4	1	flags	bit 0 END_STREAM = 1; others 0
5	1	header count	number of fields at payload start
6	2	stream ID	1-65535; 0 for connection errors
8	4	payload length	following bytes; version 1 cap 8,388,608

The magic rejects unrelated traffic; one-byte version/type fields leave extension space. A 16-bit ID suffices for sequential exchanges and permits later correlation. The four-byte length lets a receiver locate the next frame even when it cannot interpret this one; the 8 MiB cap bounds memory use. END_STREAM says this frame completes the message and leaves room for multi-frame messages later.

Forward compatibility rule

A receiver meeting a frame type it does not know **MUST read and discard exactly its payload length**, ignore that frame's flags, header count, and contents, then continue at the next fixed header. It MUST NOT parse the unknown payload as headers. An unknown type alone causes no response.

3. Payload and header names

The first **header count** fields are headers; all remaining payload bytes are the body. Indexed field: `code(1) | value length(2) | UTF-8 value`. Literal field: `00 | name length(1) | ASCII name | value length(2) | UTF-8 value`. Literal names are 1-255 bytes of lowercase letters, digits, or hyphens; values are 0-65,535 bytes. Duplicate names, unknown index codes, bad UTF-8, and lengths beyond the frame are malformed. Names are case-sensitive.

Code	Name	Sent by	Code	Name	Sent by
1	:method	client	6	:status	server
2	:path	client	7	content-length	server
3	host	client	8	content-type	server
4	user-agent	client	9	server	server
5	accept	client	10	connection	server

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Message semantics and error handling

4. Request

A request has type 1, END_STREAM set, a nonzero stream ID, and an empty body. It requires `:method` = GET, an origin-form `:path` starting with /, and `host`. The supplied client also sends `user-agent` and `accept`. A different method gets 405. Header values are UTF-8; the path's optional query after ? is ignored for file lookup.

5. Response

A response has type 2, END_STREAM set, and the matching stream ID. It includes a three-digit decimal `:status`, decimal `content-length`, `content-type`, `server`, and `connection`. Content length is the exact body byte count. A 200 body is the requested file's bytes; an error body is UTF-8 plain text. The client writes bodies to stdout, hexdumps every frame to stderr with -v, and exits nonzero for any 4xx/5xx.

6. File mapping

The server maps the decoded path beneath its configured root. It percent-decodes once and rejects invalid percent escapes, NUL, backslash, duplicate slashes, . or .. path segments, and paths that resolve outside the root (including symlinks). A directory maps to index.html. Missing or unreadable files produce 404.

7. Status and connection behavior

Status	When sent	Connection
200	File returned	stays open
400	Malformed frame/request or unsafe path	closes for framing errors; otherwise stays open
404	File missing or unreadable	stays open
405	Method other than GET	stays open

For bad magic/version/flags, an oversize or truncated payload, or malformed header encoding, the server sends 400 on stream 0 with `connection: close` if possible, then closes. A request using stream 0 also gets 400 and closes. Unknown frame types follow the skip rule on page 1. Otherwise the connection remains available for more requests. The server's idle timeout is 15 seconds; the client's is 10 seconds. Normal closure occurs between complete frames.

8. Byte-level example

For GET /index.html on stream 1, the 12-byte request prefix is `42 43 01 01 01 05 00 01 00 00 00 37`: magic BC, version 1, request type, END_STREAM, five headers, stream 1, and 55 payload bytes. The response repeats stream 1. See the accompanying annotated-hexdump.md for every byte of a complete exchange.